



Robots, Natives and Immigrants in US local labor markets[☆]

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ABSTRACT

I analyze the impact of industrial robots on the employment of natives and immigrants in US local labor markets between 1990 and 2014. The proposed mechanism, through which robot adoption affects the employment of natives and immigrants differentially, is based on two facts: first, robots tend to displace workers based on the task content of occupations, and second, natives and immigrants in the US differ in their task specialization. Therefore, robots should affect their employment unequally. Exploiting plausibly exogenous variation in robot exposure across US local labor markets over time, I test this mechanism and find that the effect on immigrants is roughly 1.76 times greater than that observed for natives. Specifically, I find that one more robot per thousand workers reduces the employment-to-population ratio of natives and immigrants by 0.38 and 0.67 percentage points, respectively. I attribute these results to the fact that immigrants specialize in jobs or tasks at risk of being automated.

1. Introduction

Rapid advances in robotics technology have led to a growing concern over the labor market impacts of robot adoption and the disparate effects it may have across different groups of workers and occupations. Recent findings suggest that robot adoption affects employment and wages significantly (Acemoglu and Restrepo, 2020; Chiacchio et al., 2018). However, the impacts of robot adoption are not borne equally among different groups of workers and occupations. For instance, studies have shown that industrial robots affect men and women differently and reduce the gender wage gap (Ge and Zhou, 2020), favor high-skilled workers as compared to low-skilled workers (Graetz and Michaels, 2018), and relate to a fall in the employment share of routine manual task-intensive occupations (de Vries et al., 2020). Although previous research on the unequal effects of industrial robots has focused on outcomes by different education groups, occupations, and gender, no single study has yet examined the differential impact of industrial robots by nativity or citizenship status.

In this paper, I examine whether the labor market shock emanating from the adoption of industrial robots differentially affects the labor market outcomes of natives and immigrants in the US between 1990 and 2014. I exploit the nativity/citizenship status distinction for two reasons. First, the fact that immigrants make up an increasingly important fraction of the US labor force itself warrants an examination of the impact of different shocks on their labor market outcomes. For

instance, the percentage of foreign-born workers participating in the US labor force was only 3.7% in 1970, but by 2019, that number had increased to 13.2%.¹ Second, there is evidence highlighting the role of immigrants in attenuating the negative impact of labor demand shocks on natives in the US (Cadena and Kovak, 2016; Yu, 2023). Thus, in this regard, it is crucial to examine whether the impact of industrial robots on natives differs after considering their effect on immigrants.

I first document that natives and immigrants in the US specialize in performing different workplace tasks. Using data from the Census, the American Community Survey (ACS), and the Occupational Information Network (O*NET), I show that immigrants specialize in routine manual tasks, whereas natives specialize in non-routine cognitive tasks. Furthermore, building on the findings of the previous literature, I argue that industrial robots are more substitutable for the job tasks in which immigrants tend to specialize. Thus, they should have a differential impact on the employment of natives and immigrants. Since native and immigrant workers are represented differentially in occupations and industries, I further complement the evidence on their task specialization by documenting their employment distribution in broad occupational groups and robot-intensive industries. I show that immigrants are more likely than their counterparts to be employed in occupations rich in routine manual tasks or in industries that use more robots.

I show considerable variation in robot adoption across industries in the US and use it to construct a measure of robot exposure at the local

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¹ Author's own calculation. Moreover, according to the US Bureau of Labor Statistics, the share of foreign-born workers in the US civilian labor force touched 17.0% in 2020.

labor market level, proxied by commuting zones (CZs) in the US. Following [Acemoglu and Restrepo \(2020\)](#), the CZ-level measure of robot exposure is constructed as a shift-share variable and defined as the sum of industry-level robot penetration weighted by each industry's baseline employment share in a CZ. Thus, the variation in robot adoption across industries exposes CZs differently depending on their industrial composition of employment. The main challenge of the analysis is that hiring workers and adopting robotics technology are deliberate choices made by a firm. Therefore, the mere correlation between robot adoption and worker outcomes would not reflect a causal effect. To overcome endogeneity issues, I instrument the US robot exposure variable using an analogous measure constructed for several European countries that are ahead of the US in robotics technology. Focusing on these countries ensures that changes in robot capital are driven by technological advances rather than trends that are idiosyncratic to the US and thus identifies the causal effect of robot adoption.

Based on the work of [Acemoglu and Restrepo \(2020\)](#), I first follow their strategy and replicate their benchmark results for the overall economy, and then modify their baseline specification to analyze the differential impact of robots on the labor market outcomes of natives and immigrants. My main finding is that robot adoption leads to differential job losses for immigrants in the US, and the employment effect for immigrants is approximately 1.76 times that observed for natives. Specifically, I find that an increase of one robot per thousand workers reduces the private employment-to-population ratio of natives and immigrants by 0.38 and 0.67 percentage points, respectively. I attribute the adverse effect that industrial robots have on the employment of immigrants to differences in the task specialization of immigrants and natives in the US. Immigrants are more likely to be employed in occupations that require performing relatively more routine manual tasks and suffer disparate job losses because these are the same jobs that robots are poised to replace. Furthermore, I break down the employment effects by occupation groups for both natives and immigrants. The main contributor to the decline in employment for both groups is identified as routine manual occupations.

Moreover, by exploiting differences in the share of immigrants across US local labor markets, I show that the effects of robots on natives' employment are worse in places with fewer immigrants. This is suggestive of immigrants mitigating the negative impact of robot adoption on natives. Concerning the effects on wages, I find that robots have the same effect on wages for both natives and immigrants. However, focusing on routine manual wages, I find that an increase of one robot per thousand workers in the robot adoption measure in a CZ is predicted to reduce the log annual wage of routine manual native and immigrant workers by 3.43% and 5.32%, respectively. I also consider the employment effects of robots on recent and established immigrants, as well as on foreign- and native-born men and women. I find that the burden of robot adoption falls disproportionately on recent immigrants and foreign-born men and women.

This paper contributes to two strands of literature. First, it adds new evidence to the literature in labor economics on the employment effects of robot adoption in the labor market. The underlying idea in this literature is that the disparate effects of robot adoption depend on the extent of complementarity or substitutability between robots and workers at different skill levels, where skills are either measured by different occupations ([de Vries et al., 2020](#)), education groups ([Graetz and Michaels, 2018](#)), gender ([Ge and Zhou, 2020](#)), or age groups ([Dauth et al., 2021](#)). The novelty of my paper is that it documents the effect of robot adoption on the labor market outcomes of natives and immigrants, adding to the body of evidence on the incidence of the displacement effects of industrial robots.

Second, it contributes to the literature on the role of immigrants in the labor market. Several studies have examined how immigrants attenuate the adverse consequences of labor demand shocks on native workers. For instance, [Cadena and Kovak \(2016\)](#) and [Basso et al. \(2019\)](#) show that the higher mobility of immigrants acts as a buffer

and attenuates the variation of native employment rates in response to the labor demand shock proceeding from the Great Recession. Alternatively, closely related to my work is the paper by [Basso et al. \(2020\)](#). Focusing on differences in task specialization between immigrants and natives in the US, they find that the specialization of immigrants in manual tasks attenuates the negative impact of computers on the routine employment of natives. Although I rely on the same idea, I examine a different technology and, thus, add important evidence to this literature by documenting how immigrants' task specialization attenuates the adverse effects of a leading automation technology – industrial robots – on native workers.

The rest of this paper is structured as follows. Section 2 outlines the theoretical considerations that link how robot adoption could affect the labor market outcomes of immigrants differentially, as well as providing evidence to support the channel identified. Section 3 describes the data sources and measurements that inform subsequent analyses. Section 4 discusses the empirical specification. Section 5 discusses the main results. Section 6 provides the conclusion.

2. Theory and descriptive evidence

2.1. Theoretical considerations

Why would robot adoption lead to differential job losses for immigrants in the US? To answer this question, it is imperative to first identify and review the evidence on the channels through which robots could affect employment in general. From a theoretical standpoint, robot adoption impacts employment through two main channels ([Acemoglu and Restrepo, 2018](#)). First, it causes a displacement effect as robots take over tasks traditionally performed by humans, and second, it leads to a productivity effect as productivity gains create new jobs and tasks elsewhere in the economy. Therefore, the overall impact on employment depends on whether the displacement or productivity effect dominates. Put differently, robot adoption will affect aggregate employment negatively in the absence of offsetting productivity effects.

The existing evidence on the employment effects of industrial robots is in line with these predictions. [Acemoglu and Restrepo \(2020\)](#) find that industrial robots affect aggregate employment negatively in the US. They attribute these effects to strong displacement forces and find no support for positive spillovers resulting from productivity gains.² On the other hand, [Dauth et al. \(2021\)](#) find no adverse effects of industrial robots on aggregate employment in Germany as workers displaced by industrial robots in the manufacturing sector reallocate to the service sector. Similarly, [de Vries et al. \(2020\)](#) examine the impact of industrial robots on the employment share of occupations across 19 industries in 37 countries. They find that increased use of industrial robots is associated with displacement effects in routine manual jobs. However, the presence of reallocation effects in non-routine analytic jobs offsets the adverse effects.

The displacement effect is most suitable to examine why industrial robots would have a differential impact on the employment of immigrants in the US, as the evidence surrounding it identifies the type of tasks that robots are poised to replace. Specifically, robots are better suited to perform routine manual tasks such as welding, assembly, handling materials, and packaging. Hence, they are more likely to affect jobs that entail performing relatively more of such tasks. For example, [Acemoglu and Restrepo \(2020\)](#) find that the negative employment effects of robots are mainly concentrated in routine manual occupations, as workers in these jobs engage in tasks that are most susceptible to being performed by robots. Similarly, [Dauth et al. \(2021\)](#) and [de Vries et al. \(2020\)](#) also find that the displacement effect is pronounced for routine manual jobs.

² The only exception is the Agriculture and Forestry sector. However, the estimates are small and imprecise.

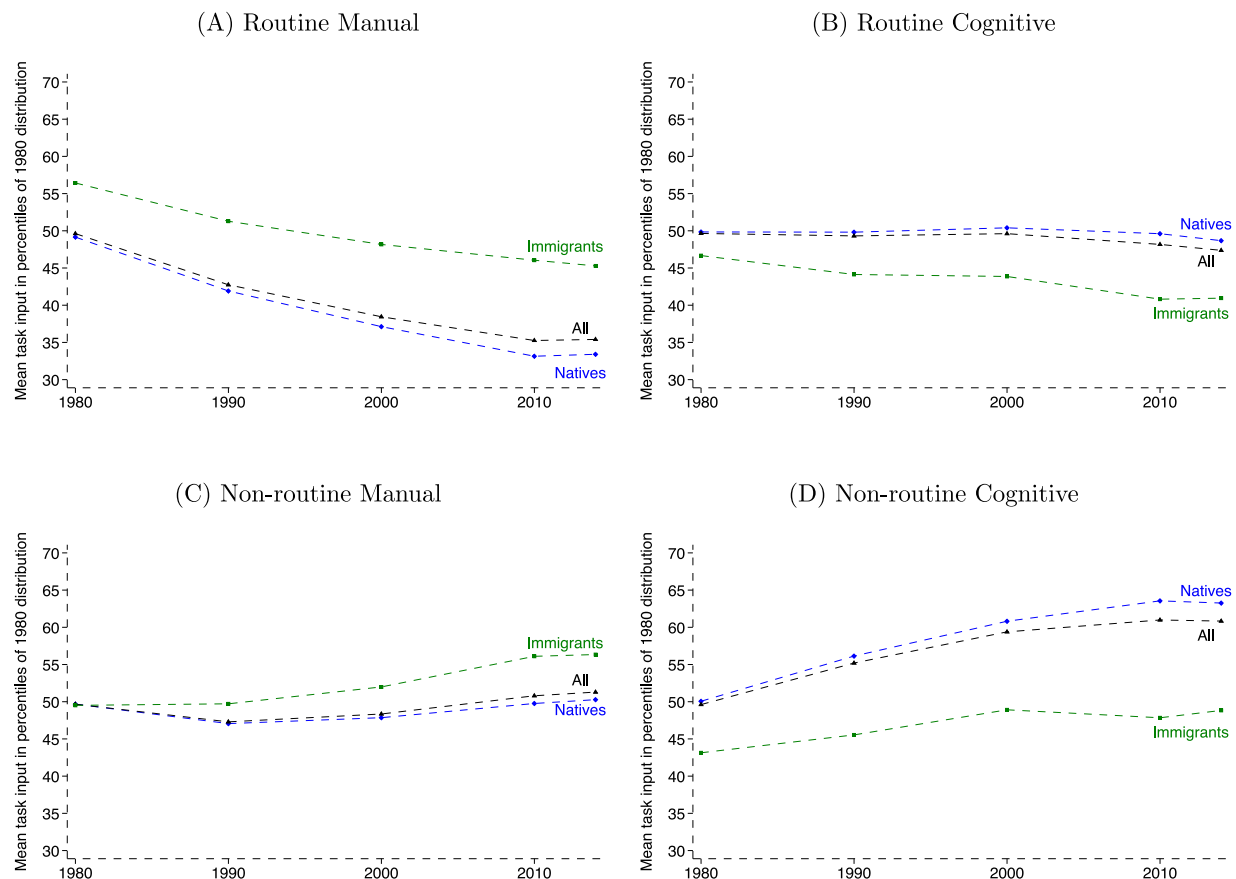


Fig. 1. Trends in Task Input in the US Economy, 1980–2014. Note: The O*NET composite task measures are paired with the 1980, 1990, and 2000 Censuses and the pooled ACS for 2009–11 and 2012–16, which represent 2010 and 2014, respectively. Changes in task inputs are expressed relative to the 1980 task distribution. The sample includes all persons aged 16–64 who report having worked last year, are civilians, and are not residing in institutional group quarters. Immigrants are defined as those who are born outside the US and are not US citizens at birth. See Online Appendix A and B.1 for details on the construction of O*NET composite task measures and consistent occupation codes, respectively.

As robots are better suited to replace workers performing routine manual tasks, one would expect that robot adoption would lead to differential job losses for immigrants only if they specialized in the jobs or tasks at risk of being automated. Interestingly, natives and immigrants in the US specialize in performing different tasks. For example, [Peri and Sparber \(2009\)](#) find that less-educated immigrants specialize in occupations requiring manual labor tasks, while their counterparts have an advantage in jobs demanding communication skills. Therefore, if I assume that immigrants specialize in occupations requiring routine manual tasks, the labor market impact of robots should vary across natives and immigrants in the US. In the analyses that follow, I provide evidence in support of this assumption using a task-based framework.

2.2. Descriptive evidence

To establish that natives and immigrants in the US specialize in performing different tasks, I begin by documenting changes in the labor input of natives and immigrants for four broad task categories between 1980 and 2014.

I adopt the two-dimensional classification of workplace tasks proposed by [Acemoglu and Autor \(2011\)](#) and distinguish between four broad task categories, namely, routine manual, routine cognitive, non-routine manual, and non-routine cognitive tasks.³ Using the Occupational Information Network (O*NET), a survey administered by the

³ The classification delineates tasks along two dimensions: routine versus non-routine, and manual versus cognitive. Routine tasks are job activities that are well-defined and involve following precise, well-understood procedures and instructions ([Autor and Price, 2013](#)) whereas non-routine tasks involve

US Department of Labor to a random sample of workers in each occupation, I construct four composite task measures that best describe the tasks performed in each of the four task categories.⁴ Finally, I assign these occupation-specific composite task measures to workers based on their reported Census or ACS occupation and average these composite task measures across workers.⁵ Following [Autor et al. \(2003\)](#), I document changes in labor input in these composite task measures in terms of their weighted percentile rank in the 1980 distribution of task inputs. [Fig. 1](#) depicts the trends in task inputs by nativity status from 1980 to 2014.

Of interest are the task input trends for routine manual tasks in Panel A of [Fig. 1](#). As can be seen, the input of routine manual tasks declined for both natives and immigrants. However, the immigrant-native input gap in routine manual tasks was, on average, 11 percentiles over the entire period, which suggests that immigrants specialize significantly more in routine manual tasks relative to natives. Furthermore, the trends in Panels C and D are consistent with the findings of [Mandelman and Zlate \(2022\)](#) and [Basso et al. \(2020\)](#), who attribute the growth

exercising discretion, using judgment, and solving problems in the case of non-routine cognitive tasks and being physically adept and responsive to unscripted interactions in the case of non-routine manual tasks ([Acemoglu and Autor, 2011](#)). See Table C1 in Online Appendix C for examples of tasks and occupations for each of the four task categories.

⁴ See Online Appendix A for details on the construction of O*NET composite task measures.

⁵ The occupations reported in the Census and the ACS are not defined consistently over time. See Online Appendix B1 for details on the construction of consistently defined occupations.

Table 1
Distribution of Employed Natives and Immigrants by Occupation Groups, 1990–2014.

	1990	2000	2014
Panel A: Natives			
Total employed (in thousands)	99135.61	107621.40	114471.40
Occupation as a percent of total employed			
Managerial, Professional, and Technical	37.87	40.97	43.65
Sales, Clerical, and Administrative Support	25.13	24.56	22.29
Production and Operative	20.88	18.19	14.95
Service	16.11	16.27	19.11
Total employed (%)	100.00	100.00	100.00
Panel B: Immigrants			
Total employed (in thousands)	9920.75	15160.14	23237.32
Occupation as a percent of total employed			
Managerial, Professional, and Technical	32.22	34.59	35.53
Sales, Clerical, and Administrative Support	18.86	17.50	14.83
Production and Operative	27.13	25.37	21.68
Service	21.79	22.53	27.97
Total employed (%)	100.00	100.00	100.00

Note: Employment shares are calculated using the 1990 and 2000 Censuses and the ACS for 2012–16. The five-year pooled ACS 2012–16 represents 2014. The sample includes all persons aged 16–64 who report being employed (excluding unpaid family workers), are civilians, and are not residing in institutional group quarters. Immigrants are defined as those who are born outside the US and are not US citizens at birth. See Online Appendix B.2 for details on the construction of four broad occupation groups.

in low-skill employment to the specialization of immigrants in manual and service jobs (non-routine manual tasks) and the growth in high-skill employment to natives' specialization in high-skilled jobs (non-routine cognitive tasks).

Using the O*NET survey to create composite task measures that best reflect the tasks performed in four broad task categories is not without limitations. As *Acemoglu and Autor (2011)* note, “O*NET contains numerous potential task scales, and it is rarely obvious which measure (if any) best represents a given task construct”. Instead, they suggest working directly with the Census and ACS occupational categories and have identified four broad occupational categories that map well into the four broad task categories. These are as follows: (1) managerial, professional, and technical occupations; (2) sales, clerical, and administrative support occupations; (3) production and operative occupations; and (4) service occupations. Therefore, I use the Census and ACS occupational categories, as the patterns depicted in *Fig. 1*, despite being informative, do not establish that immigrants in the US are more likely to be employed in routine-manual occupations (production and operative occupations) relative to natives.

To do so, I present two pieces of evidence. First, I show the distribution of employment among natives and immigrants in these four occupational groups from 1990 to 2014 in *Table 1*. When it comes to employment in production and operative occupations, both natives and immigrants witnessed a drop in their employment shares over time. However, immigrants were more likely than their counterparts to be employed in production and operative occupations in all years (for example, 22% vs. 15% in 2014). Overall, the employment shares shown in *Table 1* are consistent with a recent finding from the US Bureau of Labor Statistics, which shows that in 2021, immigrants were more likely than natives to work in service occupations and production, transportation, and material moving occupations and less likely to work in management, professional, and related occupations and sales and office occupations.⁶

Second, in *Table 2*, I show the proportions of native and immigrant workers employed in several robot-intensive industries (including automotive, basic metals, electronics, food and beverages, metal products, and plastics and chemicals) between 1990 and 2014. Among all

Table 2
Employment Shares in Robot Intensive Industries, 1990–2014.

	1990	2000	2014
All	8.96%	8.12%	6.14%
Natives	8.84%	7.94%	6.02%
Immigrants	10.09%	9.42%	6.75%

Note: Employment shares are calculated using the 1990 and 2000 Censuses and the ACS for 2012–16. The five-year pooled ACS 2012–16 represents 2014. The sample includes all persons aged 16–64 who report being employed (excluding unpaid family workers), are civilians, and are not residing in institutional group quarters. Immigrants are defined as those who are born outside the US and are not US citizens at birth. Robot-intensive industries include automotive, basic metals, electronics, food and beverages, metal products, and plastics and chemicals. See *Table 3* for robot penetration by industries.

Table 3
Penetration of Robots by Industry in the US, 1993–2014.

IFR Industries	US Penetration of Robots		
	1993–2014	1993–2000	2000–2014
<i>Manufacturing</i>			
Automotive	91.91	27.57	64.34
Miscellaneous Manufacturing	13.81	0.00	13.81
Electronics	11.60	1.61	9.99
Plastics and Chemicals	8.19	1.84	6.35
Basic Metals	7.17	0.00	7.17
Metal Products	5.59	2.89	2.70
Food and Beverages	5.05	1.19	3.86
Industrial Machinery	2.37	0.00	2.37
Minerals	0.67	0.00	0.67
Shipbuilding and Aerospace	0.54	0.00	0.54
Wood and Furniture	0.14	0.00	0.14
Paper and Printing	0.11	0.00	0.11
Textiles	0.05	0.00	0.05
<i>Non-manufacturing</i>			
Mining	0.06	0.00	0.06
Education and Research	0.06	0.00	0.06
Agriculture	0.04	0.00	0.04
Utilities	0.03	0.00	0.03
Construction	0.02	0.00	0.02
Services	0.00	0.00	0.00

Note: The penetration of robots is expressed in terms of robots per thousand workers and is defined as in Eq. (1). Data for US robot penetration in 1993 and 2000 comes from *Ge and Zhou (2020)*, which uses the 2004 distribution of industrial robots across industries to divide the total number of robots in 1993 and 2000 into IFR industries.

employed workers, the fraction working in robot-intensive industries declined from 8.96% in 1990 to 6.14% in 2014, which reflects that more labor has been displaced in robot-intensive industries. As can be seen, the fraction of immigrant workers employed in robot-intensive industries remains greater than that of native workers, although the difference seems to be shrinking over time. For instance, in 1990, 10.09% of immigrant workers were employed in robot-intensive industries, whereas for native workers, the fraction employed in the same industries was 8.84%. This suggests that immigrants are more likely than natives to be employed in robot-intensive industries, and therefore the impact of robots should vary across natives and immigrants. In fact, among native and immigrant workers, the fraction working in robot-intensive industries declined by 2.82 and 3.34 percentage points, respectively, between 1990 and 2014.

3. Data

I examine whether the labor market shock emanating from the adoption of industrial robots differentially affects the labor market outcomes of natives and immigrants in US local labor markets between 1990 and 2014. Following the previous literature (*Autor and Dorn, 2013; Acemoglu and Restrepo, 2020*), I approximate local labor markets using the concept of commuting zones (CZs) and focus on the 722 CZs that cover the mainland US. In the subsections that follow, I introduce my data sources, define my worker sample, and summarize the construction of the main variables.

⁶ See p.4 and Table 4 of the BLS report using <https://www.bls.gov/news.release/pdf/forbrn.pdf>.

3.1. Industrial robots

The most comprehensive data on the global dissemination and deliveries of industrial robots comes from the International Federation of Robotics (IFR). The IFR covers 50 countries from 1993 to 2014 and provides information on the number of industrial robots delivered by industry, country, and year. However, it is proprietary data and was obtained using the publicly available data provided by [Acemoglu and Restrepo \(2020\)](#) and [Ge and Zhou \(2020\)](#).⁷

Using these data, I define the US penetration of robots at the industry-level as:

$$PR_{i,(t_0,t_1)}^{US} = \frac{R_{i,t_1}^{US} - R_{i,t_0}^{US}}{L_{i,1990}^{US}}, \quad (1)$$

where $R_{i,t}^{US}$ represents the number of robots in industry i in the US at time t , and $L_{i,1990}^{US}$ is the baseline employment level in industry i . I compute this measure for two time periods, 1993–2000 and 2000–2014, as well as the long difference between 1993–2014, and present the trends in [Table 3](#). As can be seen, the penetration of robots is not even across industries in the US. There were few robots in the US during 1993–2000. The automotive industry had the highest number of industrial robots, followed by metal products, plastics and chemicals, electronics, and food and beverages. The average penetration of robots was 35 robots per thousand workers between 1993 and 2000. However, it accelerated after 2000. Almost every industry witnessed a significant change in robot intensity, with the average penetration of robots being 112 robots per thousand workers between 2000 and 2014. Overall, robot intensity grew to 147 robots per thousand workers across US industries when considering the entire period.

3.2. Local labor market data

3.2.1. Commuting zones

I approximate local labor markets using the concept of CZs and focus on the 722 CZs covering the mainland US. My analysis draws on the Integrated Public Use Micro Samples (IPUMS) of the US Census for the years 1970, 1990, and 2000 and the American Community Survey (ACS). To increase the sample size, I follow [Acemoglu and Restrepo \(2020\)](#) and measure the 2007 and 2014 outcomes using the ACS for 2006–2008 and 2012–2016, respectively. I use the crosswalk files provided by David Dorn to assign Census/ACS geographic units (county groups or Public Use Micro Areas (PUMAs)) to CZs.

3.2.2. Sample

My worker sample consists of individuals aged 16 and above who are not residing in institutional group quarters (such as prisons and psychiatric institutions). I define immigrants as those who are born outside the US and are not US citizens at birth. Moreover, I define recent immigrants as those who arrived in the US within the last ten years and established immigrants as those who arrived more than ten years ago.

⁷ Daron Acemoglu has made the cleaned version of IFR data publicly available at <https://economics.mit.edu/faculty/acemoglu/data/robotsjobs>. The raw IFR data has three limitations. First, around 30% of industrial robots are not assigned to any of the 19 IFR industries. Second, it reports only the overall stock of robots for North America as a whole. Finally, it does not provide an industry breakdown of robots in the US until 2004. To calculate industry-level US robot penetration in 1993 and 2000, I use data from [Ge and Zhou \(2020\)](#), which uses the 2004 distribution of industrial robots across industries to divide the total number of robots in 1993 and 2000 into IFR industries. See [Acemoglu and Restrepo \(2020\)](#) and [Ge and Zhou \(2020\)](#) for details on addressing the shortcomings in the raw IFR data.

3.2.3. Labor market outcomes

My outcomes of interest are the change in private employment-to-population ratio and the change in the occupational employment-to-population ratio of natives and immigrants in each CZ. As in [Acemoglu and Restrepo \(2020\)](#), I use the private employment-to-population ratio as the main indicator for examining employment outcomes in my analysis. This measure focuses on salaried workers exclusively within the private sector and thus excludes individuals employed in the public sector or engaged in self-employment activities. For the analysis of occupational employment, I focus on four broad occupational groups, namely, routine manual, routine cognitive, non-routine manual, and non-routine cognitive occupations. Examining employment changes within occupational groups requires constructing a consistent panel of occupation codes that cover the 1990 and 2000 Censuses and the ACS 2012–2016. To do so, I redefine the Census/ACS occupational classification system for every decennial Census and the ACS. I use the updated *occ1990dd* system, constructed by [Deming \(2017\)](#), and extend it further to cover the occupations in the 2012–2016 ACS.⁸ I end up with a version of the *occ1990dd* system that contains 329 consistently defined occupations for the period 1990–2016.⁹ I also examine changes in the real log annual wages (indexed to 2014 dollars) of natives and immigrants in each CZ. In the wage sample, I introduce further sample restrictions and focus specifically on individuals in the private sector who worked a positive number of hours and weeks in the previous year and who received a strictly positive wage income.

3.2.4. US exposure to robots

Ideally, had the IFR data provided geographical information on the within-US distribution of industrial robots at the CZ level, the best measure of a CZ's exposure to robots would be constructed by dividing the number of robots in a CZ by its baseline employment. However, the IFR reports data on the number of industrial robots at the country-by-industry level, which does not allow the construction of a metric of robot exposure at the CZ level. Therefore, following [Acemoglu and Restrepo \(2020\)](#), I measure a CZ's exposure to robots using a shift-share approach and define it as the sum of industry-level robot penetration weighted by each industry's baseline employment share in the CZ.¹⁰ More specifically, it is constructed as follows:

$$US \text{ exposure to robots}_{c,(t_0,t_1)} = \sum_i \frac{L_{i,c,1990}}{L_{c,1990}} \times PR_{i,(t_0,t_1)}^{US}, \quad (2)$$

where $\frac{L_{i,c,1990}}{L_{c,1990}}$ is the employment share of industry i in CZ c in 1990 and $PR_{i,(t_0,t_1)}^{US}$ is defined as in Eq. (1). In other words, I use a shift in the national increase in robots in a given industry weighted by the local employment share of that industry.

[Fig. 2](#) depicts the geographic distribution of robot exposure in the US between 1993 and 2014, with darker shades signifying higher levels of exposure. Many CZs have only a modest increase of 0.18–0.52 robots per thousand workers. In others, the figure ranges from

⁸ [Autor and Dorn \(2013\)](#) created the original *occ1990dd* system, which included 330 consistently defined occupations for the 1980, 1990, and 2000 Censuses and the 2005 ACS. [Deming \(2017\)](#) extended the system to cover the 2010–2011 ACS occupation codes.

⁹ See Appendix B for details on the construction of consistent occupation codes and broad occupation groups.

¹⁰ The measure used in [Acemoglu and Restrepo \(2020\)](#) adjusts the $PR_{i,(t_0,t_1)}^{US}$ term in Eq. (1) by accounting for the overall expansion of industry-specific output. This is done by combining IFR industry codes with output by industry data from the European Union-level analysis of capital, labor, energy, materials, and service inputs (EU KLEMS) Growth and Productivity Accounts. As I do not have access to raw IFR data, I am unable to adjust the $PR_{i,(t_0,t_1)}^{US}$ term in Eq. (1) to account for the overall expansion of industry-specific output. However, as discussed in [Acemoglu and Restrepo \(2020\)](#), as well as in [Ge and Zhou \(2020\)](#), the adjustment term is not quantitatively important due to the low number of robots in 1993.

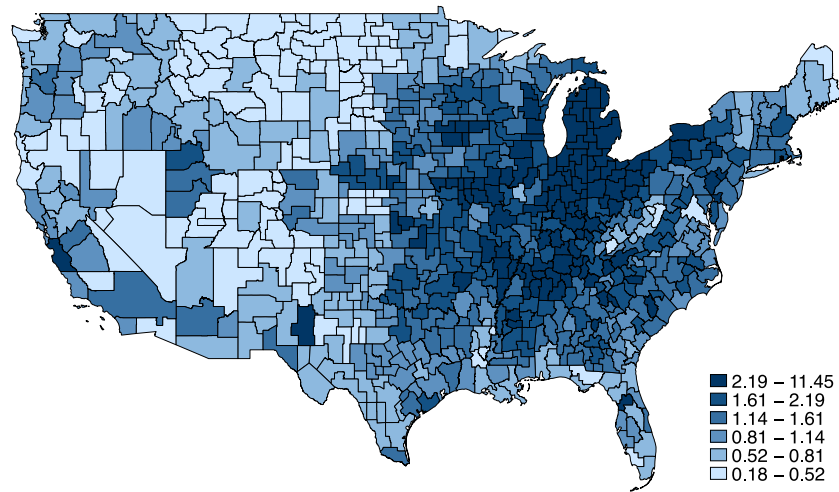


Fig. 2. Geographic Distribution of US Exposure to Robots, 1993–2014. Note: The figure depicts the geographic distribution of US robot exposure (in robots per thousand workers) at the CZ level between 1993 and 2014. US exposure to robots in a CZ is defined as the sum of industry-level robot adoption weighted by each industry's 1990 employment share in the CZ.

0.81 to 2.19 robots per thousand workers. As industrial robots are mainly employed in a subset of manufacturing industries, the majority of CZs with the highest robot adoption (> 2.19 robots per thousand workers) are situated in the Rust Belt (Michigan, Ohio, and Indiana), which is known as the manufacturing hub of the US. Overall, there is a substantial variation in the penetration of industrial robots across CZs and over time, which I exploit in my analysis.

3.2.5. Other shocks

To control for potentially confounding changes in trade patterns and other technological advances, I use data on exposure to Chinese imports from Autor et al. (2019) and data on the fraction of employment in a CZ in routine occupations from Autor and Dorn (2013), respectively. Controlling for these variables is important as skill-biased technical change and Chinese import competition have impacted the evolution of employment and wage structure in the US since the 1980s.

4. Empirical strategy

To identify the effects of industrial robots on the labor market outcomes of natives and immigrants, I use a stacked differences specification with 722 CZs and two time periods: 1990–2000 and 2000–2014.¹¹ The estimating equation, which is estimated separately for natives and immigrants, is given by:

$$\Delta Y_{c,t} = \beta_1 \cdot US \text{ exposure to robots}_{c,t} + \beta_2 \cdot X'_{c,t} + \gamma_t + \epsilon_{c,t}, \quad (3)$$

where $\Delta Y_{c,t}$ is the change in an outcome variable, such as the private employment-to-population ratio or the routine manual employment-to-population ratio, of natives or immigrants in CZ c between years t_0 and t_1 , and $US \text{ exposure to robots}_{c,t}$ is the change in CZ c 's exposure to robots between years t_0 and t_1 .¹² This equation stacks the two time periods covering the interval 1990–2014 and includes time dummies for each decade (γ_t) as well as various other CZ characteristics and economic variables that may affect the labor market outcomes of

natives and immigrants. The control vector $X'_{c,t}$ includes start-of-decade values of several covariates at the CZ-level to allow for differential trends by baseline demographic and industry characteristics, as well as changes in trade patterns and other technological advances.¹³ Finally, $\epsilon_{c,t}$ represents an idiosyncratic error term.

Even after taking into account all of the control variables in vector X , estimating Eq. (3) using ordinary least squares (OLS) could lead to biased estimates as the variable US exposure to robots could be correlated with the error term $\epsilon_{c,t}$. It might be the case that variation in US exposure to robots is driven by CZ-specific factors that also influence labor market outcomes. For example, positive demand shocks may induce firms to adopt robots and simultaneously impact their labor demand, thus leading to a positive bias in the estimates of the effect of robots.

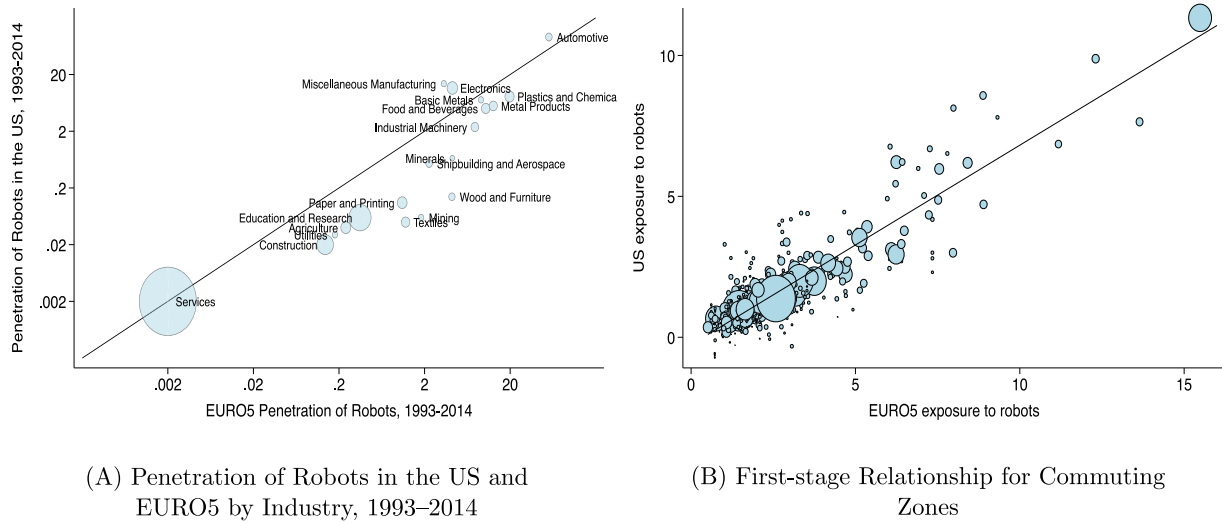
To overcome such endogeneity issues, as in Acemoglu and Restrepo (2020), I instrument the US exposure to robots variable using a similar measure constructed for five European countries that are ahead of the US in robotics technology. Focusing on countries that are ahead of the curve in robotics technology allows us to abstract from trends that are idiosyncratic to the US and, thus, remove the influence of confounding factors. Consider any robot-intensive industry in the US as an example. Using US data would capture all the ups and downs the industry might have witnessed. As a result, a strategy like this enables me to isolate such variation and identify the component of robot penetration that is solely driven by exogenous improvements in technology.¹⁴

¹¹ The stacked differences specification allows for exploiting the accelerated robot penetration, which started in the 2000s. Outcomes for the 1990–2000 period are adjusted to a 7-year equivalent change for consistency since the IFR data starts in 1993.

¹² The occupational employment-to-population ratios are defined as $\frac{Employment_{c,t}^{N,O}}{Population_{c,t}^{N,O}}$, where N represents the nativity status, and O represents employment in routine manual, non-routine manual, routine cognitive, or routine cognitive occupations.

¹³ Baseline CZ-level demographic and industry covariates are constructed using 1990 and 2000 US Census data. I include the following demographic characteristics: log population; share of women; share of immigrants; share of population over 65 years old; shares of population with no college, some college, college, and a master's or doctoral degree; and population shares of Whites, Blacks, Hispanics, and Asians. I include the following industry characteristics: share of employment in manufacturing and light manufacturing; female share of manufacturing employment; and immigrant share of manufacturing employment. I control for changes in trade patterns and other technological advances using two variables: exposure to Chinese imports and the share of employment in routine jobs (see Section 3.2.5).

¹⁴ For a strategy like this to work, it is critical to identify countries that are not just ahead of the US in robotics technology but also where the evolution of US robot adoption closely tracks the average of selected countries. In particular, as shown in Figure C1 in Online Appendix C, the number of robots per thousand workers in Denmark, Finland, France, Italy, and Sweden (EURO5) matches this criteria quite well. Thus, tracing out what is happening in these EURO5 countries serves as a proxy for what the US is following.



(A) Penetration of Robots in the US and EURO5 by Industry, 1993–2014

(B) First-stage Relationship for Commuting Zones

Fig. 3. Industry Correlations and the First-stage Relationship. Note: *Panel A:* Plot of the EURO5 penetration of robots and the US penetration of robots between 1993 and 2014. The penetration of robots is given in terms of the number of robots per thousand workers in the industry. The solid line corresponds to the 45° line, and the circle size indicates the baseline US employment in the industry. *Panel B:* The figure presents the relationship between EURO5 exposure to robots and US exposure to robots for the period 1993–2014. Covariates (baseline demographic and industry characteristics, as well as changes in trade patterns and other technological advances) are partialled out. The solid line corresponds to a long-differences regression with the CZ population in 1990 as weights. Marker size indicates the 1990 population in the CZ.

Specifically, I construct the instrumental variable (IV) as:

$$EURO5 \text{ exposure to robots}_{c,t_0,t_1} = \sum_i \frac{L_{i,c,1970}}{L_{c,1970}} \times \frac{1}{5} \sum_{j \in EURO5} \frac{M_{i,t_1}^j - M_{i,t_0}^j}{L_{i,1990}^j}, \quad (4)$$

where $M_{i,t}^j$ represents the number of robots in industry i in country j at time t , $L_{i,1990}^j$ represents the baseline employment level in industry i in and country j , and $\frac{L_{i,c,1970}}{L_{c,1970}}$ is the employment share of industry i in CZ c in 1970.¹⁵ EURO5 countries include Denmark, Finland, France, Italy, and Sweden. In other words, I instrument the shift component of Eq. (2) using industry-level robot penetration in countries ahead of the US in robotics technology. In addition, I instrument the share component with plausibly exogenous local employment shares from 1970 to focus on differences in the industrial specialization of CZs that predate robotics technology (Acemoglu and Restrepo, 2020; Lerch, 2020).

The IV relies on two assumptions: first, that robot adoption trends in the US are driven by those in European countries, and second, that the adoption of robots in European countries is unrelated to conditions in US local labor markets. To test the first condition, I document industry correlations in the US and EURO5 countries in Panel A of Fig. 3. As can be seen, there is a positive correlation between the adoption of robots in the US and EURO5 countries. Although Panel A of Fig. 3 depicts a close association between the industry-level spread of robots in the US and EURO5 countries, I further confirm the relevance of the instrument by depicting the first-stage relationship in Panel B of Fig. 3. As can be seen, US exposure to robots is strongly correlated with robot exposure in the EURO5, which is consistent with the notion that US robot adoption is driven by technological advances in the EURO5. Moreover, with a first-stage F -statistic of 221, the instrument passes conventional thresholds for strong instruments.

As far as the exclusion restriction is concerned, one cannot completely rule out the possibility that the adoption of robots in European countries is unrelated to conditions in local labor markets in the US.

However, Acemoglu and Restrepo (2020) have shown that the penetration of robots in EURO5 does not mimic other industry-level trends. In particular, the industries that are adopting robots more are not those that are affected by competition from Chinese or Mexican imports, offshoring, rapid growth in total capital or IT capital, or that have a large fraction of routine jobs.

5. Results

5.1. Employment effects

As my empirical strategy closely follows that of Acemoglu and Restrepo (2020), it is crucial to begin by highlighting the deviations I introduce when replicating and extending their work. There are two notable deviations. First, I do not adjust my explanatory variable (see Eq. (2)) for the overall expansion of industry-specific output, which differs from their approach. This is due to the unavailability of raw IFR data for my analysis.¹⁶ However, as discussed in Acemoglu and Restrepo (2020), as well as in Ge and Zhou (2020), the adjustment term is not quantitatively important due to the low number of robots in 1993. Moreover, I construct the instrumental variable, EURO5 exposure to robots, both with and without the adjustment term, which ensures that this deviation does not impact my reduced-form estimates. Second, in addition to the covariates used in Acemoglu and Restrepo (2020), I include two more covariates: the share of immigrants in the population and the immigrant share of manufacturing employment.

Building upon this foundation, I proceed to replicate Acemoglu and Restrepo's (2020) benchmark employment effects for the overall economy during the period 1990–2007 and subsequently extend the analysis to examine the effects separately for natives and immigrants during the same period.¹⁷ To do this, I estimate the reduced-form

¹⁵ The measure used in Acemoglu and Restrepo (2020) adjusts the EURO5 robot penetration term in Eq. (4) for the overall expansion of industry-specific output as well. However, as discussed in footnote 10, the adjustment term is not quantitatively important.

¹⁶ Instead, to calculate industry-level US robot penetration (see Eq. (1)) for 1993 and 2000, I rely on data from Ge and Zhou (2020). Unfortunately, this dataset cannot be combined with the data on output at the industry level from EUKLEMS Growth and Productivity Accounts.

¹⁷ I undertake the replication of Table 3 (columns 1 and 4 in Panel A) and Online Appendix Table A24 (column 4 in Panel F) as presented in Acemoglu and Restrepo's (2020) study. The former uses the EURO5 exposure to robots variable with the adjustment term, whereas the latter does so without the

Table 4

Effects of Robots on Employment: Stacked Differences 1990–2000 and 2000–2007, Reduced-form Estimates. (Dependent variable: Change in private employment-to-population ratio)

	With adjustment term				Without adjustment term			
	Weighted by population			Unweighted	Weighted by population			Unweighted
	(1)	(2)	(3)		(5)	(6)	(7)	
Panel A: All								
EURO5 exposure to robots	−0.63*** (0.09)	−0.56*** (0.06)	−0.56*** (0.05)	−0.72*** (0.09)	−0.48*** (0.09)	−0.41*** (0.05)	−0.41*** (0.05)	−0.52*** (0.09)
Observations	1444	1444	1444	1444	1444	1444	1444	1444
R ²	0.25	0.42	0.42	0.40	0.22	0.40	0.40	0.38
Panel B: Natives								
EURO5 exposure to robots	−0.52*** (0.09)	−0.49*** (0.05)	−0.49*** (0.05)	−0.69*** (0.09)	−0.41*** (0.10)	−0.38*** (0.05)	−0.38*** (0.05)	−0.49*** (0.09)
Observations	1444	1444	1444	1444	1444	1444	1444	1444
R ²	0.20	0.50	0.50	0.44	0.19	0.48	0.48	0.43
Panel C: Immigrants								
EURO5 exposure to robots	−0.78*** (0.29)	−0.77*** (0.28)	−0.81*** (0.26)	−1.22*** (0.45)	−0.53** (0.26)	−0.51* (0.26)	−0.57** (0.24)	−0.98** (0.39)
Observations	1444	1444	1444	1444	1444	1444	1444	1444
R ²	0.26	0.30	0.31	0.12	0.25	0.30	0.30	0.11
Covariates								
Time dummies and census divisions	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Baseline controls	No	Yes	Yes	Yes	No	Yes	Yes	Yes
Additional controls	No	No	Yes	Yes	No	No	Yes	Yes

Note: $N = 1444$ (722 CZs times 2 time periods). Robust standard errors in parentheses are clustered on the state level. Columns 1–3 and 5–7 present regressions weighted by population in 1990. Columns 4 and 8 present unweighted regressions. The covariates included in each specification are reported at the bottom of the table. Columns 1 and 5 include only census division dummies and time period dummies. Columns 2 and 6 add the following controls: log population; share of women; share of population over 65 years old; shares of population with no college, some college, college, and a master's or doctoral degree; population shares of Whites, Blacks, Hispanics, and Asians; share of employment in manufacturing and light manufacturing; female share of manufacturing employment; exposure to Chinese imports; and the share of employment in routine jobs. Columns 3 and 7 add the following controls: the share of immigrants and the immigrant share of manufacturing employment. *Significant at the 10 percent level. **Significant at the 5 percent level. ***Significant at the 1 percent level.

specification of Eq. (3), where I regress changes in employment on the EURO5 exposure to robots variable, looking at both scenarios with and without the adjustment term. Table 4 presents the results of this replication, with columns 1–4 providing estimates for the adjusted EURO5 exposure to robots variable and columns 5–8 providing estimates for the unadjusted EURO5 exposure to robots variable. The table examines three outcome variables: the change in private employment-to-population ratio for all individuals in Panel A, for natives in Panel B, and for immigrants in Panel C. Columns 1–3 and columns 5–7 show estimates weighted by population in 1990, whereas columns 4 and 8 present unweighted estimates. Standard errors are clustered at the state level to account for spatial correlation across CZs.

Within Panel A, columns 1 and 5 provide estimates of a parsimonious specification that includes only geographic dummies for the nine Census divisions and time dummies as covariates. These columns show a strong negative association between overall employment changes and adjusted as well as unadjusted EURO5 exposure to robots, respectively. These estimates show that an increase of one robot per thousand workers in the adjusted and unadjusted robot exposure measures within a CZ is associated with a predicted reduction of 0.63 and 0.48 percentage points, respectively, in the private employment-to-population ratio for all individuals. These estimates have to be interpreted carefully since omitted variables could confound the effects of industrial robots. Thus, to account for potential confounders, I incorporate a complete set of covariates associated with the regional and economic characteristics of CZs, as used in Acemoglu and Restrepo (2020), into the remaining columns.

Columns 2 and 6 in Panel A include the following start-of-period demographic and industry characteristics of CZs: log population; share of women; share of population over 65 years old; shares of population

with no college, some college, college, and a master's or doctoral degree; population shares of Whites, Blacks, Hispanics, and Asians; share of employment in manufacturing and light manufacturing; and female share of manufacturing employment. Additionally, I control for potentially confounding factors arising from shifts in trade patterns and technological advancements. This is captured by variables such as import competition from China and the share of employment in routine jobs or occupations. As can be seen in columns 2 and 6 in Panel A, adding these covariates reduces the estimates slightly. For instance, the private employment-to-population ratio for all individuals falls to 0.56 and 0.41 percentage points, respectively.

While columns 2 and 6 stand as the preferred specifications in Acemoglu and Restrepo's (2020) work, I use an alternative specification shown in columns 3 and 7. Within these columns, I add two additional covariates. The first covariate, the share of immigrants in the population, accounts for the enclave effect. A large immigrant network could depress robot adoption by providing a source of cheap labor while also affecting labor market outcomes for both immigrants and natives. Hence, not accounting for this could lead to biased estimates. The second covariate, the immigrant share of manufacturing employment, addresses the sector-specific adoption of industrial robots. Given that industrial robots are predominantly used in manufacturing, variations in the demographic composition of this workforce could confound the observed effects. The inclusion of these controls does not affect the coefficients of interest pertaining to overall employment.

Lastly, I conduct additional robustness checks. In Panel A, columns 4 and 8 demonstrate that the employment effects remain statistically significant even in unweighted regressions. Furthermore, Online Appendix Table C2, Panel A, columns 1 and 3, exclude the top 1% of CZs with the highest robot exposure, illustrating that these specific CZs are not accountable for the estimates. Similarly, in Panel A of Online Appendix Table C2, columns 2 and 4, the EURO5 exposure to robots variable is constructed without considering the automotive industry. The presence of significant negative estimates implies that the automotive industry is not the driving factor behind these estimates.

adjustment term. The results of this replication correspond to columns 1 and 2 in Panel A, and column 6 in Panel A, of Table 4, respectively.

Table 5

Effects of Robots on Employment: Stacked Differences 1990–2000 and 2000–2014, Reduced-form Estimates. (Dependent variable: Change in private employment-to-population ratio)

	With adjustment term				Without adjustment term			
	Weighted by population			Unweighted	Weighted by population			Unweighted
	(1)	(2)	(3)		(5)	(6)	(7)	
Panel A: All								
EURO5 exposure to robots	−0.57*** (0.17)	−0.44*** (0.12)	−0.44*** (0.12)	−0.75*** (0.17)	−0.44*** (0.14)	−0.31*** (0.09)	−0.31*** (0.08)	−0.56*** (0.13)
Observations	1444	1444	1444	1444	1444	1444	1444	1444
R ²	0.19	0.33	0.33	0.43	0.18	0.32	0.32	0.42
Panel B: Natives								
EURO5 exposure to robots	−0.47*** (0.13)	−0.38*** (0.10)	−0.38*** (0.10)	−0.71*** (0.16)	−0.37*** (0.12)	−0.28*** (0.07)	−0.27*** (0.07)	−0.53*** (0.13)
Observations	1444	1444	1444	1444	1444	1444	1444	1444
R ²	0.16	0.32	0.32	0.47	0.15	0.31	0.31	0.46
Panel C: Immigrants								
EURO5 exposure to robots	−0.72** (0.27)	−0.66** (0.27)	−0.71*** (0.26)	−1.46*** (0.49)	−0.54** (0.25)	−0.42* (0.22)	−0.49** (0.21)	−1.11** (0.39)
Observations	1444	1444	1444	1444	1444	1444	1444	1444
R ²	0.26	0.32	0.33	0.20	0.26	0.31	0.32	0.20
Covariates								
Time dummies and census divisions	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Baseline controls	No	Yes	Yes	Yes	No	Yes	Yes	Yes
Additional controls	No	No	Yes	Yes	No	No	Yes	Yes

Note: $N = 1444$ (722 CZs times 2 time periods). Robust standard errors in parentheses are clustered on the state level. Columns 1–3 and 5–7 present regressions weighted by population in 1990. Columns 4 and 8 present unweighted regressions. The covariates included in each specification are reported at the bottom of the table. Columns 1 and 5 include only census division dummies and time period dummies. Columns 2 and 6 add the following controls: log population; share of women; share of population over 65 years old; shares of population with no college, some college, college, and a master's or doctoral degree; population shares of Whites, Blacks, Hispanics, and Asians; share of employment in manufacturing and light manufacturing; female share of manufacturing employment; exposure to Chinese imports; and the share of employment in routine jobs. Columns 3 and 7 add the following controls: the share of immigrants and the immigrant share of manufacturing employment. *Significant at the 10 percent level. **Significant at the 5 percent level. ***Significant at the 1 percent level.

Before expanding the analysis to investigate effects on natives and immigrants during 1990–2007, it is crucial to establish a baseline by comparing my estimates with those of [Acemoglu and Restrepo \(2020\)](#). My estimates in columns 1 and 2 in Panel A and column 6 in Panel A of [Table 4](#) replicate the results from [Table 3](#) (columns 1 and 4 in Panel A) and Online Appendix Table A24 (column 4 in Panel F) in [Acemoglu and Restrepo's \(2020\)](#) study. Notably, my estimates closely align and are always within two ninths of a standard error of those reported in their work. This comparison lays the foundation for understanding the effects on both natives and immigrants.

In Panels B and C of [Table 4](#), I present estimates of the impact of robots on the employment of natives and immigrants from 1990 to 2007. In conclusion, whether I use adjusted or unadjusted measures of robot exposure and whether I look at my preferred specifications (columns 3 and 7) or [Acemoglu and Restrepo's \(2020\)](#) preferred specifications (columns 2 and 6), the results show that industrial robots hurt immigrants more than their counterparts. Even when looking at unweighted regressions (columns 4 and 8 in [Table 4](#)) and excluding the top 1% of CZs with the highest robot exposure or constructing the EURO5 exposure to robots variable without taking into account the automotive industry (Panels B and C of Online Appendix Table C2), these findings stay the same.

So far, I have shown benchmark employment effects from 1990 to 2007 for the overall economy, natives, and immigrants using reduced-form estimates. [Table 5](#) extends this analysis to 1990–2014.¹⁸ Comparing the outcomes of [Tables 4](#) and [5](#) provides two intriguing insights. Firstly, when examining my preferred specifications (columns 3 and 6) across both tables for overall employment as well as employment among natives and immigrants, the estimates for the 1990–2014 period show a slight decrease compared to the 1990–2007 period. This trend persists whether adjusted or unadjusted measures are used. Secondly,

immigrants experience substantially more adverse impacts in both tables compared to their counterparts. A noteworthy observation is that when assessing relative effects within my preferred specifications, a distinct pattern emerges. Specifically, [Table 5](#) highlights more pronounced relative effects. To elaborate, while [Table 4](#) reveals immigrants' effects as roughly 1.5 times greater than those for natives, [Table 5](#) demonstrates effects approximately 1.8 times greater.

The inability to adjust my explanatory variable for the overall expansion of industry-specific output means that the 2SLS estimates I provide are derived from unadjusted robot penetration measures. Nevertheless, it is worth noting that excluding the adjustment term would not significantly impact my 2SLS estimates. This is because the relative magnitude of the employment effects for immigrants and natives in [Table 5](#) is approximately similar. Specifically, the employment effect for immigrants is 1.86 times greater than that observed for natives with the adjustment term and 1.8 times greater without it.

To conclude and quantify the aggregate implications of robot adoption on the employment of natives and immigrants, I show in [Table 6](#) the 2SLS estimates of the effects of industrial robots on the employment of natives and immigrants from 1990 to 2014. The coefficients presented in the table are estimates of β_1 from Eq. (3). I stack observations for natives and immigrants and allow all independent variables to have separate coefficients for natives and for immigrants by interacting each right-hand-side variable with a dummy for nativity status. The p -value is from a test for the equality of coefficients. I instrument robot adoption in the US by the average change in the five European countries as described in Eq. (4). The first-stage estimates suggest that the instrument has a positive and significant effect on US exposure to robots. The large F -statistics reflect that the IV has high explanatory power for the variable of the interest.

Once again, I find a negative impact for both natives and immigrants, with the effect being more adverse for immigrants. For example, in column 3, the estimates imply that an increase of one robot per thousand workers in exposure to robots is associated with a decline in natives' private employment-to-population ratio of 0.38 percentage

¹⁸ Robustness checks accompanying [Table 5](#) are presented in Online Appendix Table C3.

Table 6

Effects of Robots on Employment: Stacked Differences 1990–2000 and 2000–2014, 2SLS Estimates. (Dependent variable: Change in private employment-to-population ratio)

	Weighted			Unweighted
	(1)	(2)	(3)	(4)
Natives	–0.51*** (0.18)	–0.38*** (0.11)	–0.38*** (0.11)	–0.89*** (0.25)
Immigrants	–0.74** (0.35)	–0.58* (0.32)	–0.67** (0.31)	–1.85*** (0.67)
Test for equality of coefficients (<i>p</i> -value)	0.28	0.39	0.19	0.06
First-stage coefficient	0.73*** (0.03)	0.73*** (0.04)	0.72*** (0.03)	0.60*** (0.06)
First-stage <i>F</i> -statistic	243.31	317.89	325.77	49.65
Covariates				
Time dummies and census divisions	Yes	Yes	Yes	Yes
Baseline controls	No	Yes	Yes	Yes
Additional controls	No	No	Yes	Yes

Note: The number of observations for each regression is 2888 (722 CZs $\times$ 2 years $\times$ two groups). Robust standard errors in parentheses are clustered on the state level. Columns 1–3 present regressions weighted by population in 1990. Column 4 presents unweighted regressions. The covariates included in each specification are reported at the bottom of the table. Column 1 includes only census division dummies and time period dummies. Column 2 adds the following controls: log population; share of women; share of population over 65 years old; shares of population with no college, some college, college, and a master's or doctoral degree; population shares of Whites, Blacks, Hispanics, and Asians; share of employment in manufacturing and light manufacturing; female share of manufacturing employment; exposure to Chinese imports; and the share of employment in routine jobs. Column 3 adds the following controls: the share of immigrants and the immigrant share of manufacturing employment. *Significant at the 10 percent level. **Significant at the 5 percent level. ***Significant at the 1 percent level.

points and a fall of 0.67 percentage points for immigrants. Notably, the standard error for natives stands at 0.11, while for immigrants, it is 0.31, indicating a larger dispersion in the latter group's estimates. The *t*-tests to evaluate coefficient equality fail to reject the null hypothesis at conventional levels in the weighted regressions, with a *p*-value of 0.19 in my preferred specification in column 3. In the unweighted specification, column 4, the difference in coefficients is statistically significant with a *p*-value of 0.06.

While the precision of the estimated parameters could thus be better, there is still overall a reasonable basis to argue that immigrants are more adversely affected by robots compared to natives, based on the point estimates and the consistent trend in the results. Moreover, it is important to consider the possibility that these estimates may not fully capture the negative impact on immigrant employment rates. Should displaced immigrants opt to leave the US, a scenario more likely for them compared to their counterparts, this could potentially lead to an underestimation of the negative impact on immigrant employment in absolute terms.

5.2. Employment effects by occupation

The results in Section 5.1 support my hypothesis that robot adoption will lead to differential job losses for immigrants in the US. In this subsection, I extend my analysis to a decomposition of estimates across five broad occupational groups.¹⁹ This is crucial as it not only provides a more detailed insight into the varying vulnerabilities within the labor market but also unveils potential patterns of displacement and adaptation across different skill sets. Moreover, decomposing the impacts by occupation groups helps shed light on the heterogeneous skill responses.

Table 7 presents the 2SLS estimates for the effects of robots on the occupational employment of natives and immigrants, respectively. Focusing on employment in routine manual occupations first, I find a negative impact for both natives and immigrants, with the effect being more adverse for immigrants. For example, the estimates in column 3 imply that an increase of one robot per thousand workers in exposure to robots is associated with a decline in natives' routine manual employment-to-population ratio of 0.19 percentage points and a fall

of 0.44 percentage points for immigrants. These results underscore a significant trend, revealing that the primary source of employment decline for both natives and immigrants stems from routine manual occupations. One potential explanation for the more pronounced effect on immigrants could be that, even within the group of routine manual occupations, immigrants are particularly likely to work in those occupations that are more susceptible to robotization. Further investigation supports this, revealing that immigrants in routine manual occupations are more highly concentrated in robot-intensive industries than natives, thereby increasing their susceptibility.

Before examining the remaining occupational groups, it is important to note that, as opposed to other countries, the absence of positive spillovers from robot adoption in the US makes it an outlier and leads to robots affecting aggregate employment negatively. This anomaly implies that the remaining job losses for natives and immigrants would be observed in either non-routine cognitive, routine cognitive, non-routine manual, or agricultural occupations. As in [Acemoglu and Restrepo \(2020\)](#), I find no support for positive employment effects in either of the remaining four occupation groups (except non-routine manual employment for natives in column 3). Interestingly, for natives, I find that the job losses are concentrated in non-routine cognitive occupations, with a point estimate of –0.14 percentage points. On the other hand, the effects for immigrants are found in non-routine manual occupations, though the magnitude of these effects is considerably smaller than that reported for employment losses in routine manual jobs (–0.11 percentage points vs. –0.44 percentage points). Though explaining the contrasting impacts of robots on non-routine manual occupations for immigrants and natives appears difficult, natives' decline in non-routine cognitive employment appears to be in line with the idea posed by [Deming \(2017\)](#). That is, automation technologies might be substituting for labor at higher levels in the skill distribution.

5.3. Effects by non-employment

Given that neither natives nor immigrants have experienced positive employment spillovers, I now turn my attention to examining how robots influence their rates of unemployment and non-participation in [Table 8](#). I estimate a positive impact of robots on the non-participation and unemployment rates of both natives and immigrants. The estimates in column 3 imply that around eight-tenths of the additional non-employed natives stop participating in the labor force, while two-tenths remain unemployed. For immigrants, the estimates in column 3 imply

¹⁹ Estimates for agricultural occupations are presented in Online Appendix Table C4.

Table 7

Effects of Robots on Occupational Employment: Stacked Differences 1990–2000 and 2000–2014, 2SLS Estimates. (Dependent variable: Change in occupational employment-to-population ratio)

	Weighted			Unweighted
	(1)	(2)	(3)	(4)
Panel A: Routine manual				
Natives	−0.28*** (0.10)	−0.19*** (0.05)	−0.19*** (0.04)	−0.47*** (0.12)
Immigrants	−0.43* (0.22)	−0.34 (0.22)	−0.44** (0.20)	−1.16*** (0.41)
Test for equality of coefficients (<i>p</i> -value)	0.40	0.38	0.13	0.05
Panel B: Non-routine cognitive				
Natives	−0.18*** (0.06)	−0.14*** (0.04)	−0.14*** (0.04)	−0.27*** (0.07)
Immigrants	−0.07 (0.13)	−0.08 (0.09)	−0.09 (0.09)	−0.19 (0.26)
Test for equality of coefficients (<i>p</i> -value)	0.36	0.50	0.55	0.75
Panel C: Routine cognitive				
Natives	−0.09* (0.05)	−0.08** (0.04)	−0.08* (0.04)	−0.20** (0.09)
Immigrants	−0.05 (0.04)	−0.03 (0.04)	−0.02 (0.04)	−0.05 (0.11)
Test for equality of coefficients (<i>p</i> -value)	0.27	0.28	0.20	0.28
Panel D: Non-routine manual				
Natives	0.03*** (0.01)	0.03*** (0.01)	0.03** (0.01)	0.03 (0.03)
Immigrants	−0.18** (0.08)	−0.12** (0.06)	−0.11* (0.06)	−0.36 (0.28)
Test for equality of coefficients (<i>p</i> -value)	0.00	0.02	0.04	0.16
Covariates				
Time dummies and census divisions	Yes	Yes	Yes	Yes
Baseline controls	No	Yes	Yes	Yes
Additional controls	No	No	Yes	Yes

Note: Panel E of Table 7 is shown in Online Appendix Table C4. The number of observations for each regression is 2888 (722 CZs × 2 years × two groups). Robust standard errors in parentheses are clustered on the state level. Columns 1–3 present regressions weighted by population in 1990. Column 4 presents unweighted regressions. The covariates included in each specification are reported at the bottom of the table. Column 1 includes only census division dummies and time period dummies. Column 2 adds the following controls: log population; share of women; share of population over 65 years old; shares of population with no college, some college, college, and a master's or doctoral degree; population shares of Whites, Blacks, Hispanics, and Asians; share of employment in manufacturing and light manufacturing; female share of manufacturing employment; exposure to Chinese imports; and the share of employment in routine jobs. Column 3 adds the following controls: the share of immigrants and the immigrant share of manufacturing employment. *Significant at the 10 percent level. **Significant at the 5 percent level. ***Significant at the 1 percent level.

that around nine-tenths drop out of the labor force, while one-tenth remain unemployed.

Importantly, the potential for bias in the coefficient estimates, which arises from the possibility of displaced immigrants leaving the US, also applies to these outcomes. In such a scenario, immigrants would not appear in the unemployment or non-participation data, possibly leading to an underestimation of the positive impacts on these rates in absolute terms. However, it is worth noting that the differences in these estimated coefficients are not statistically significant, making the interpretation of these results, at best, suggestive.

Overall, these findings shed light on the significant effect of robots on labor market dynamics. Notably, the higher non-participation rates among immigrants hold implications for their integration into the labor market, emphasizing the need for targeted support. Additionally, the implications extend to natives as well, emphasizing the importance of implementing strategies such as language and skills training, job placement services, and other support mechanisms to aid successful reintegration into the labor market and society at large.

5.4. Results for wages

I present the effects of robots on the wages of natives and immigrants in Table 9. Based on the employment effects, one would also expect the effect of robots on the wages of immigrants to be greater than that of natives. This line of reasoning follows Ge and Zhou (2020) and posits that if the only contribution of robots was to replace routine manual tasks, it should hurt immigrants' wages more as they specialize

in the same tasks. Hence, I test this hypothesis by estimating the impact of robots on the wages of natives and immigrants, respectively.

Table 9 reports the 2SLS estimates for the effects of industrial robots on the annual wages of natives and immigrants between 1990 and 2014. More specifically, I regress the change in average real log annual wages of natives and immigrants on US robot adoption, respectively. Focusing on my preferred specification, column 3, I find that an increase of one robot per thousand workers in the robot adoption measure in a CZ is predicted to reduce the average log annual wage of natives and immigrants by 2.60% and 2.68%, respectively.

The wage effects on natives and immigrants are essentially the same, and despite the fact that the effects run counter to the reasoning from Ge and Zhou (2020), they can potentially be explained by emphasizing that they could be caused by the negative employment effects found in non-routine cognitive occupations for natives. Non-routine cognitive occupations necessitate a high level of education and pay well. On the other hand, the differential employment effects for immigrants are concentrated in low-paying routine manual and non-routine manual occupations. As the dependent variable is the change in average real log annual wages, it is plausible that it might be capturing these countervailing employment effects.

To circumvent this issue, I also regress the change in average real log annual wages of routine manual native and immigrant workers on US robot adoption in Table 9, respectively. Focusing on my preferred specification, column 3, I find that an increase of one robot per thousand workers in the robot adoption measure in a CZ is predicted to reduce the log annual wage of routine manual native and immigrant

Table 8

Effects of Robots on Unemployment and Non-participation: Stacked Differences 1990–2000 and 2000–2014, 2SLS Estimates. (Dependent variables: Change in unemployment-to-population ratio and change in non-participation ratio)

	Weighted			Unweighted
	(1)	(2)	(3)	(4)
Panel A: Unemployment rate				
Natives	0.10** (0.04)	0.11*** (0.03)	0.11*** (0.03)	0.19*** (0.06)
Immigrants	0.10** (0.05)	0.07* (0.03)	0.07* (0.04)	0.26 (0.16)
Test for equality of coefficients (<i>p</i> -value)	0.94	0.36	0.38	0.66
Panel B: Non-participation rate				
Natives	0.57*** (0.16)	0.44*** (0.09)	0.43*** (0.08)	0.92*** (0.21)
Immigrants	0.71** (0.35)	0.57* (0.33)	0.68** (0.30)	1.11* (0.60)
Test for equality of coefficients (<i>p</i> -value)	0.63	0.61	0.29	0.72
Covariates				
Time dummies and census divisions	Yes	Yes	Yes	Yes
Baseline controls	No	Yes	Yes	Yes
Additional controls	No	No	Yes	Yes

Note: The number of observations for each regression is 2888 (722 CZs × 2 years × two groups). Robust standard errors in parentheses are clustered on the state level. Columns 1–3 present regressions weighted by population in 1990. Column 4 presents unweighted regressions. The covariates included in each specification are reported at the bottom of the table. Column 1 includes only census division dummies and time period dummies. Column 2 adds the following controls: log population; share of women; share of population over 65 years old; shares of population with no college, some college, college, and a master's or doctoral degree; population shares of Whites, Blacks, Hispanics, and Asians; share of employment in manufacturing and light manufacturing; female share of manufacturing employment; exposure to Chinese imports; and the share of employment in routine jobs. Column 3 adds the following controls: the share of immigrants and the immigrant share of manufacturing employment. *Significant at the 10 percent level. **Significant at the 5 percent level. ***Significant at the 1 percent level.

Table 9

Effects of Robots on Wages: Stacked Differences 1990–2000 and 2000–2014, 2SLS Estimates. (Dependent variables: Change in average real log annual wages and change in average real log annual wages of routine manual occupations)

	Weighted			Unweighted
	(1)	(2)	(3)	(4)
Panel A: All occupations				
Natives	−2.75*** (0.44)	−2.57*** (0.26)	−2.60*** (0.29)	−5.05*** (0.88)
Immigrants	−3.35*** (1.21)	−2.58** (1.05)	−2.68** (1.06)	−5.09*** (1.68)
Test for equality of coefficients (<i>p</i> -value)	0.50	0.99	0.92	0.98
Panel B: Routine manual occupations				
Natives	−3.49*** (0.64)	−3.36*** (0.43)	−3.43*** (0.45)	−6.93*** (1.05)
Immigrants	−6.17*** (1.38)	−5.37*** (1.09)	−5.32*** (1.07)	−9.65*** (3.18)
Test for equality of coefficients (<i>p</i> -value)	0.01	0.02	0.02	0.33
Covariates				
Time dummies and census divisions	Yes	Yes	Yes	Yes
Baseline controls	No	Yes	Yes	Yes
Additional controls	No	No	Yes	Yes

Note: The number of observations for each regression is 2888 (722 CZs × 2 years × two groups). Robust standard errors in parentheses are clustered on the state level. Columns 1–3 present regressions weighted by population in 1990. Column 4 presents unweighted regressions. The covariates included in each specification are reported at the bottom of the table. Column 1 includes only census division dummies and time period dummies. Column 2 adds the following controls: log population; share of women; share of population over 65 years old; shares of population with no college, some college, college, and a master's or doctoral degree; population shares of Whites, Blacks, Hispanics, and Asians; share of employment in manufacturing and light manufacturing; female share of manufacturing employment; exposure to Chinese imports; and the share of employment in routine jobs. Column 3 adds the following controls: the share of immigrants and the immigrant share of manufacturing employment. *Significant at the 10 percent level. **Significant at the 5 percent level. ***Significant at the 1 percent level.

workers by 3.43% and 5.32%, respectively. This refined measure might potentially circumvent, to some extent, the compositional effects discussed above, providing a more nuanced understanding of the wage effects.

5.5. The mitigating effects of immigrants

So far, I have investigated whether the effects of industrial robots on natives and immigrants are identical or distinct. However, this does not mean that immigrants mitigate the negative effects of robots on natives. To show that the effects on natives would have been worse if there were fewer immigrants, I exploit differences in the share of

immigrants across US local labor markets to examine whether CZs with fewer immigrants experience more pronounced negative effects on native employment.

Table 10 presents the 2SLS estimates for the effects of robots on the employment of natives in CZs with a high and low share of immigrants. CZs are classified into four groups based on the shares of immigrants in the CZ at the start of the decade, that is, in 1990 and 2000. Particularly, I distinguish between CZs with a top 25% and bottom 25% share of immigrants and those with a top 33% and bottom 33% share of immigrants. Focusing on my preferred specifications in column 3, I find that an increase of one robot per thousand workers in the robot

Table 10

Effects of Robots on Employment of Natives in Commuting Zones with High and Low Share of Immigrants: Stacked Differences 1990–2000 and 2000–2014, 2SLS Estimates. (Dependent variable: Change in private employment-to-population ratio of natives)

	Weighted			Unweighted
	(1)	(2)	(3)	(4)
Panel A: Top vs. Bottom 25% immigrant share				
Top 25%	–0.42*** (0.15)	–0.25*** (0.07)	–0.24*** (0.09)	–0.80 (0.44)
Bottom 25%	–0.61*** (0.14)	–0.33*** (0.12)	–0.32*** (0.11)	–0.65*** (0.22)
Test for equality of coefficients (<i>p</i> -value)	0.37	0.52	0.55	0.79
Panel B: Top vs. Bottom 33% immigrant share				
Top 33%	–0.43*** (0.16)	–0.32*** (0.07)	–0.31*** (0.08)	–0.80** (0.37)
Bottom 33%	–0.69*** (0.16)	–0.35*** (0.11)	–0.34*** (0.12)	–0.71*** (0.26)
Test for equality of coefficients (<i>p</i> -value)	0.25	0.83	0.81	0.86
Covariates				
Time dummies and census divisions	Yes	Yes	Yes	Yes
Baseline controls	No	Yes	Yes	Yes
Additional controls	No	No	Yes	Yes

Note: CZs are characterized into the four groups based on the start-of-decade share of immigrants in a CZ. Panel A has 722 and Panel B has 962 observations. Robust standard errors in parentheses are clustered on the state level. Columns 1–3 present regressions weighted by population in 1990. Column 4 presents unweighted regressions. The covariates included in each specification are reported at the bottom of the table. Column 1 includes only census division dummies and time period dummies. Column 2 adds the following controls: log population; share of women; share of population over 65 years old; shares of population with no college, some college, college, and a master's or doctoral degree; population shares of Whites, Blacks, Hispanics, and Asians; share of employment in manufacturing and light manufacturing; female share of manufacturing employment; exposure to Chinese imports; and the share of employment in routine jobs. Column 3 adds the following controls: the share of immigrants and the immigrant share of manufacturing employment. *Significant at the 10 percent level. **Significant at the 5 percent level. ***Significant at the 1 percent level.

adoption measure in CZs with the lowest 25% share of immigrants is expected to lower the employment-to-population ratio of natives by 0.32 percentage points and by 0.24 percentage points in CZs with the highest 25% share of immigrants. The results paint a similar picture of worse outcomes for natives in CZs with the lowest 33% share of immigrants and CZs with the highest 33% share of immigrants (0.34 percentage points vs. 0.31 percentage points). However, it is important to note that the differences in these estimated coefficients are not statistically significant at conventional levels, thus underscoring the need for caution when interpreting these results.

While estimating the causal impact of the mitigating effect provided by immigrants is beyond the scope of this paper, the underlying mechanism that could be at play relates to the mobility of immigrants (see Yu (2023)). Being the most mobile workforce, immigrants may have moved towards CZs less affected by robot adoption. The fact that I find less negative outcomes for natives in CZs with a high share of immigrants could be due to a larger labor supply adjustment following the higher outmigration of immigrants, which could only be possible in CZs with a high share of immigrants.

5.6. Employment effects by immigrants' duration of stay and gender

In this subsection, I examine the impact of robots on the employment of recent and established immigrants, as well as foreign- and native-born men and women. I start by looking at how robot adoption affects the employment of recent immigrants (those who arrived in the US within the last ten years) and established immigrants (those who arrived more than ten years ago). Distinguishing immigrants by their duration of stay is important for two reasons. First, due to the convergence of labor market outcomes between immigrants and natives. For instance, recent literature mostly concludes that the immigrant-native gap in labor market outcomes diminishes over the duration of stay (Kerr and Kerr, 2011). Second, as first documented by Peri and Sparber (2009), there are marked differences in the supply of tasks by recent and established immigrants in the US.

Before presenting the employment results, I document the trends in routine manual tasks of recent and established immigrants in Fig. 4 using the framework outlined in Section 2.2. Fig. 4 depicts that recent immigrants specialize significantly more in routine manual tasks

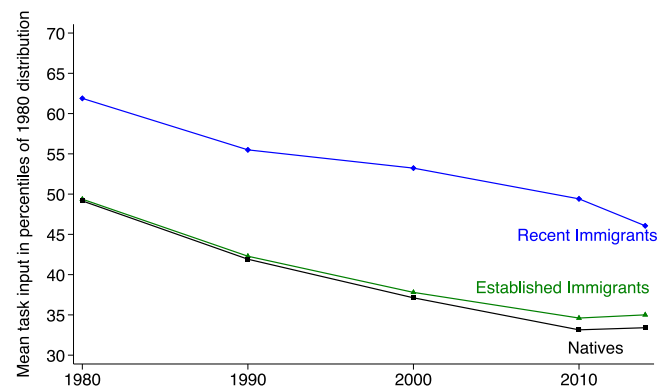


Fig. 4. Trends in Routine Manual Task Input, 1980–2014. Note: The O*NET composite task measures are paired with the 1980, 1990, and 2000 Censuses and the pooled ACS for 2009–11 and 2012–16, which represent 2010 and 2014, respectively. Changes in task inputs are expressed relative to the 1980 task distribution. Recent immigrants are defined as those who arrived in the US within the last ten years, and established immigrants are those who arrived more than ten years ago. See Online Appendix A and B.1 for details on the construction of O*NET composite task measures and consistent occupation codes, respectively.

relative to established immigrants. Interestingly, when compared with the trends shown in Panel A of Fig. 1 (the black line in Fig. 4), it appears that the supply of routine manual tasks by established immigrants coincides with that of the natives. Hence, I expect robots to predominantly hurt the employment of recent immigrants, and the impact on established immigrants will be of a similar magnitude as that observed for natives.

I test this line of reasoning by estimating the impact of robots on the employment of recent and established immigrants, respectively. Fig. 5 depicts the 2SLS estimates for the effects of robots on the employment of recent immigrants, established immigrants, and natives between 1990 and 2014. The results are consistent with the reasoning stated above. For example, the point estimates in my preferred specification (with additional controls) are indicative of two things. First, the burden of robot adoption falls predominantly on recent immigrants. The

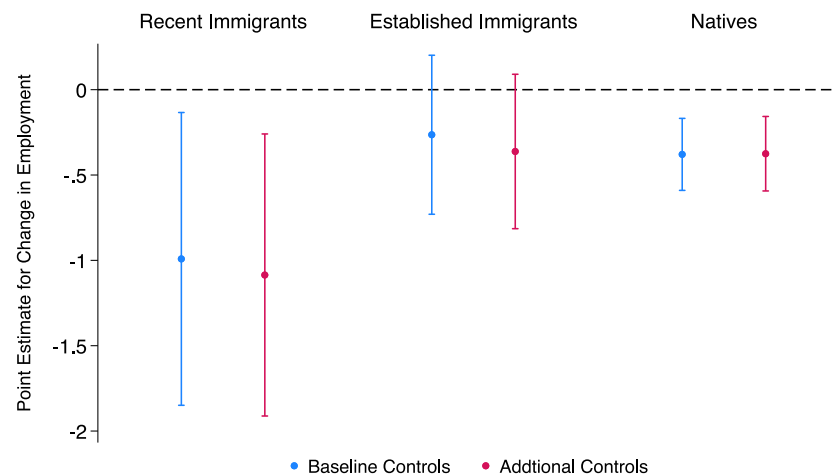


Fig. 5. Effects of Robots on Employment of Recent and Established Immigrants: Stacked Differences 1990–2000 and 2000–2014, 2SLS Estimates. Note: This figure presents 2SLS estimates of the effects of robots on changes in private employment-to-population ratios of recent immigrants, established immigrants and natives. The capped lines provide 95% confidence intervals. The 2SLS estimates correspond to the specification as in columns 2 and 3 of Table 6.

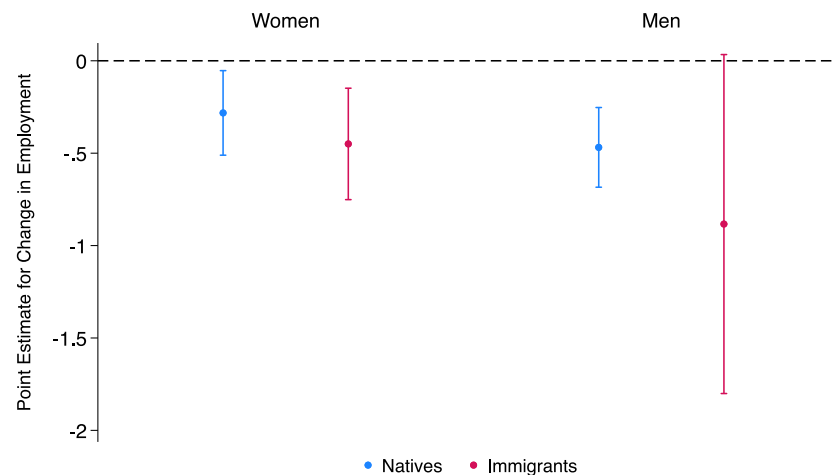


Fig. 6. Effects of Robots on Employment by Gender and Nativity Status: Stacked Differences 1990–2000 and 2000–2014, 2SLS Estimates. Note: This figure presents 2SLS estimates of the effects of robots on changes in private employment-to-population ratios of natives and immigrants by gender. The capped lines provide 95% confidence intervals. The 2SLS estimates correspond to the specification as in column 3 of Table 6.

estimates imply that an increase of one robot per thousand workers in the robot adoption measure in a CZ is predicted to reduce the employment of recent and established immigrants by 1.09 and 0.36 percentage points, respectively. However, the latter estimate is statistically insignificant. Second, the impact on established immigrants appears to closely mirror the magnitude observed for natives (−0.36 and −0.38 percentage points, respectively).

Turning now to the effect of robots on the employment of foreign- and native-born men and women, I start by highlighting the underlying mechanism by which robots can affect their employment disproportionately. A priori, as noted by Ge and Zhou (2020), robot capital is more substitutable for brawn skills, in which men have a comparative advantage. Therefore, I expect robots to affect men disproportionately. Fig. 6, which depicts the estimates of the impact of robots on employment by gender and nativity status, confirms this. I find negative effects for both men and women; however, the effects on men are more pronounced. For instance, the point estimates for foreign-born men and women are −0.88 and −0.44, respectively.²⁰ For native-born

men and women, they are −0.47 and −0.28, respectively.²¹ Regarding the effects being larger for immigrants, I resort to the main hypothesis advanced in this paper and rely on the findings from the US Bureau of Labor Statistics' Labor Force Characteristics of Foreign-Born Workers reports, which consistently show that foreign-born men and women are more likely to be employed in routine manual occupations than their counterparts.

6. Conclusion

This paper analyzes the impact of industrial robots on the labor market outcomes of natives and immigrants in US local labor markets between 1990 and 2014 and investigates whether robot adoption leads to differential job losses for immigrants in the US. Exploiting plausibly exogenous variation in robot adoption across labor markets and over time, I show that robot adoption differentially affects the employment of natives and immigrants, with the effect for immigrants being 1.76 times the size of that observed for natives. Specifically, my results suggest that an additional robot per thousand workers reduces the

²⁰ The estimates are significant at the 10% and 1% levels, respectively.

²¹ The estimates are significant at the 1% and 5% levels, respectively.

private employment-to-population ratio of natives and immigrants by 0.38 and 0.67 percentage points, respectively. To explain the more adverse impact on the employment of immigrants, I exploit differences in task specialization between natives and immigrants and argue that immigrants' specialization in the jobs or tasks most susceptible to being replaced by industrial robots leads them to suffer accordingly.

The implications of these results are far-reaching, underscoring the heightened susceptibility of immigrants' jobs to automation. In light of the rising share of foreign-born workers in the US civilian labor force (sitting at 17% in 2020), it is a policy-relevant issue with profound implications since, on the one hand, rapid advances in automation technologies in the US could threaten immigrant integration in the future, and on the other hand, admitting more immigrants is considered to play an important role in mitigating symptoms of the US economy's long-run malaise, such as low productivity growth and declining domestic geographic mobility (see [Orrenius et al. \(2020\)](#)). While there is evidence suggesting that low-skilled immigration encourages native workers to move into skilled occupations through training (see [Mandelman and Zlate \(2022\)](#)), the retraining and reskilling of immigrants in the age of automation provides an opportunity to harness and optimize the potential of the immigrant workforce in the United States, as well as to assist immigrants in acquiring the appropriate skills for an economy that is undergoing rapid transformation.

Data availability

Data will be made available on request.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.labeco.2023.102456>.

References

- Acemoglu, Daron, Autor, David, 2011. Skills, tasks and technologies: Implications for employment and earnings. In: *Handbook of Labor Economics*. Elsevier, pp. 1043–1171.
- Acemoglu, Daron, Restrepo, Pascual, 2018. The race between man and machine: Implications of technology for growth, factor shares, and employment. *Amer. Econ. Rev.* 108 (6), 1488–1542.
- Acemoglu, Daron, Restrepo, Pascual, 2020. Robots and jobs: Evidence from US labor markets. *J. Polit. Econ.* 128 (6), 2188–2244.
- Autor, David H., Dorn, David, 2013. The growth of low-skill service jobs and the polarization of the US labor market. *Amer. Econ. Rev.* 103 (5), 1553–1597.
- Autor, David, Dorn, David, Hanson, Gordon, 2019. When work disappears: Manufacturing decline and the falling marriage market value of Young men. *Am. Econ. Rev.: Insights* 1 (2), 161–178.
- Autor, D.H., Levy, F., Murnane, R.J., 2003. The skill content of recent technological change: An empirical exploration. *Q. J. Econ.* 118 (4), 1279–1333.
- Autor, David, Price, Brendan, 2013. The Changing Task Composition of the US Labor Market: An Update of Autor, Levy, and Murnane (2003). Working Paper.
- Basso, Gaetano, D'Amuri, Francesco, Peri, Giovanni, 2019. Immigrants, labor market dynamics and adjustment to shocks in the Euro area. *IMF Econ. Rev.* 67 (3), 528–572.
- Basso, Gaetano, Peri, Giovanni, Rahman, Ahmed S., 2020. Computerization and immigration: Theory and evidence from the United States. *Canad. J. Econ./Rev. Canad. D'écon.* 53 (4), 1457–1494.
- Cadena, Brian C., Kovak, Brian K., 2016. Immigrants equilibrate local labor markets: Evidence from the Great Recession. *Am. Econ. J.: Appl. Econ.* 8 (1), 257–290.
- Chiacchio, Francesco, Petropoulos, Georgios, Pichler, David, 2018. The Impact of Industrial Robots on EU Employment and Wages: A Local Labour Market Approach. Bruegel Working Paper.
- Dauth, Wolfgang, Findeisen, Sebastian, Suedekum, Jens, Woessner, Nicole, 2021. The adjustment of labor markets to robots. *J. Eur. Econom. Assoc.* 19 (6), 3104–3153.
- de Vries, Gaaitzen J., Gentile, Elisabetta, Miroudot, Sébastien, Wacker, Konstantin M., 2020. The rise of robots and the fall of routine jobs. *Labour Econ.* 66, 101885.
- Deming, David J., 2017. The growing importance of social skills in the labor market. *Q. J. Econ.* 132 (4), 1593–1640.
- Ge, Suqin, Zhou, Yu, 2020. Robots, computers, and the gender wage gap. *J. Econ. Behav. Organ.* 178, 194–222.
- Graetz, Georg, Michaels, Guy, 2018. Robots at work. *Rev. Econ. Stat.* 100 (5), 753–768.
- Kerr, Sari Pekkala, Kerr, William R., 2011. Economic impacts of immigration: A survey. *Finn. Econ. Pap.* 24 (1), 1–32.
- Lerch, Benjamin, 2020. Robots and nonparticipation in the US: Where have all the displaced workers gone? SSRN Electron. J..
- Mandelman, Federico S., Zlate, Andrei, 2022. Offshoring, automation, low-skilled immigration, and labor market polarization. *Am. Econ. J.: Macroecon.* 14 (1), 355–389.
- Orrenius, Pia M., Zavodny, Madeline, and, Stephanie Gullo, 2020. How Does Immigration Fit into the Future of the U.S. Labor Market?. Federal Reserve Bank of Dallas, Working Papers, 2020, Federal Reserve Bank of Dallas.
- Peri, Giovanni, Sparber, Chad, 2009. Task specialization, immigration, and wages. *Am. Econ. J.: Appl. Econ.* 1 (3), 135–169.
- Yu, Chan, 2023. The role of immigrants in the United States labor market and Chinese import competition. *J. Int. Econ.* 144, 103792.